

SAMPLE QUESTION PAPER

IB DIPLOMA
CHEMISTRY

1. The alloy bronze contains:
- Copper and zinc
 - Copper and tin
 - Copper, tin and zinc
 - Copper and magnesium

ANSWER	
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2. Rusting of iron requires
- air
 - water
 - air & water
 - none of the above

ANSWER	
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3. Which one of the following is an Arrhenius base?
- ammonia
 - sodium hydroxide
 - sodium carbonate
 - potassium carbonate

ANSWER	
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4. Which one of the following elements reacts most vigorously with water?
- Iron
 - Potassium
 - Calcium
 - Silver

ANSWER	
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5. The molecular formula of ethanal is
- HCHO
 - CH₃CH₂OH
 - CH₃CHO
 - CH₃COOH

ANSWER	
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SECTION B

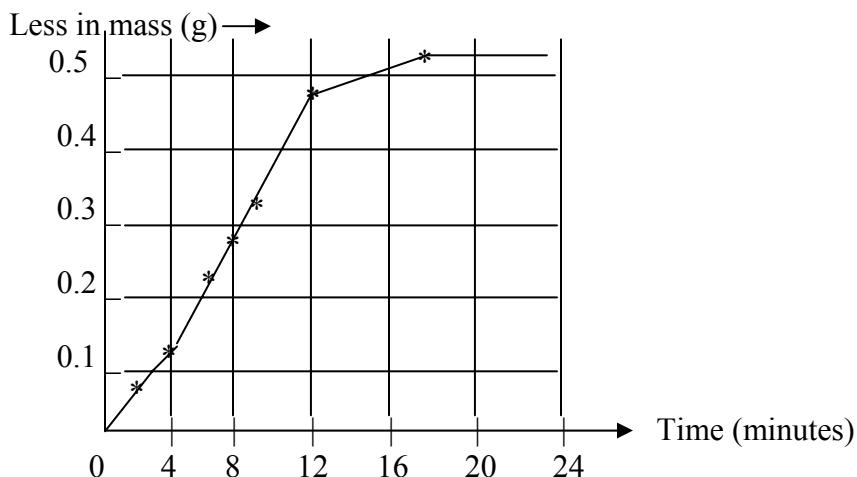
6. Ethanoic acid is used for making synthetic vinegar and ethanol is used as an industrial solvent and as a sterilizing agent.
- a) Give the equation for the reaction between ethanol and ethanoic acid in the presence of concentrated sulphuric acid. What is the reaction known as?

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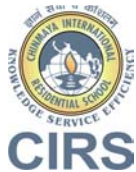
- What happens when ethanoic acid is heated with sodalime? Give the equation. What is the reaction known as?
- Give a test to distinguish between ethanol and ethanoic acid.
- What is denatured alcohol? What is the purpose of denaturing alcohol?

SECTION C

- The following information concerns an experiment to investigate the rate of a reaction between sodium carbonate and excess of sulphuric acid. 1.136 g of sodium carbonate were put into 10 ml of 10^{-1} M sulphuric acid in a small flask. The mass of the flask and its contents were determined at two-minute intervals to determine the loss of mass caused by the evolution of carbon dioxide. The results were recorded in the form of a graph given below. In this graph, the loss of mass is plotted against time. Use the graph to answer the questions, which follow:



- What is the mass of carbon dioxide formed in 8 minutes from the start of the reaction?
- State two means by which the production of carbon dioxide can be increased.
- What was the maximum mass of carbon dioxide formed?
- Explain why the loss of mass recorded during the 18th, 20th and 22nd minutes remains the same.
- Write an equation for the reaction between sodium carbonate and sulphuric acid.



CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL, COIMBATORE

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SECTION D

8. Methane is the major constituent in CNG. The complete combustion of methane produces carbondioxide.
- a) Write the chemical equation for the complete combustion of methane.
 - b) Calculate the number of molecules of methane present in 48 g of methane.
(atomic mass of C=12g, H=1g) (Avogadro number = 6.023×10^{23})
 - c) Draw the electron cross-dot diagram for a methane molecule.
 - d) Methane is a saturated hydrocarbon. Explain what do you mean by saturated hydrocarbon